Unit 7

GEOMETRY

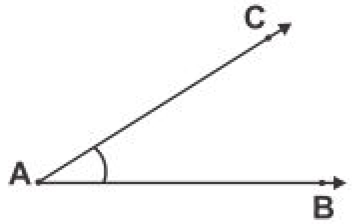
7.1 ANGLES

7.1.1 Recall an angle and recognize acute, right, obtuse, straight and reflex angle.

We know when two rays $\overrightarrow{AB}$ and $\overrightarrow{AC}$ meet at common end point, they form an angle.

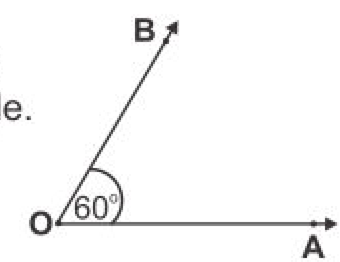
Point A is called its vertex. $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are called its arms or sides.

The angle at point A is named as angle BAC or angle CAB.
"$\angle$" is the symbol for an angle.
It is written as $\angle$ BAC or $\angle$ CAB.
It is read as angle BAC or angle CAB.

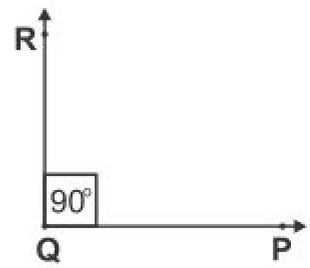


7.1.1 Kinds of an Angle

(i) Acute angle:
An angle whose measure is greater than 0° but less than 90° is called an acute angle.
Here $m\angle AOB = 60^\circ$
(m stands for measure)
Therefore $\angle$ AOB is an acute angle.



(ii) Right angle:
An angle whose measure is 90°, is called right angle.
It mkes a perfect 'L' shape.
Here $m\angle PQR = 90^\circ$
Therefore $\angle$ PQR is a right angle.



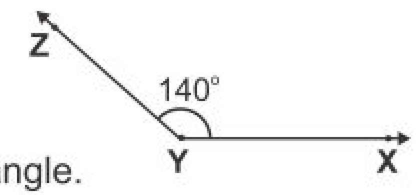
Teacher's Note
Teacher should revise and recall the students acute, right, obtuse, straight and reflex angle through making with paper or rope.

(iii) Obtuse angle:

An angle whose measure is greater than 90°but less than 180° is called an obtuse angle.

Here $m\angle XYZ = 140°$

Therefore $\angle XYZ$ is an obtuse angle.

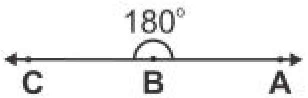


(iv) Straight angle:

An angle whose measure is 180°is called a straight angle.

Here $m\angle ABC = 180°$

Therefore $\angle ABC$ is a straight angle.

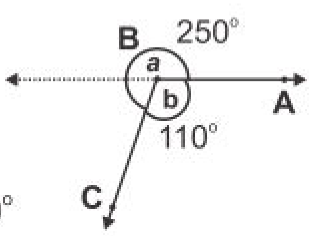


(v) Reflex angle:

An angle whose measure is greater than 180°but less than 360°is called a reflex angle.

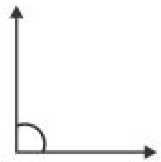
In the given figure $m\angle ABC = a = 250°$

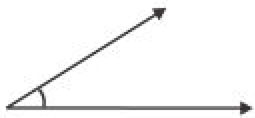
Therefore $\angle ABC$ is a reflex angle.

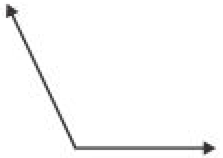


EXERCISE 7.1

1. Name the angles then write their measurements.

(a) 

(b) 

(c) 

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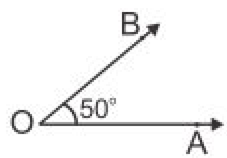
2. Identify the angles as acute, right, obtuse, straight or reflex angle.

(a)



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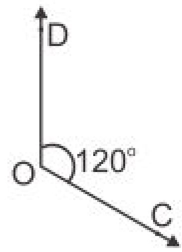

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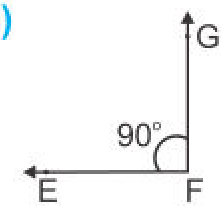

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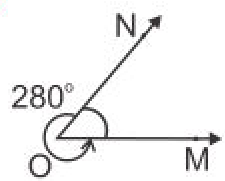

(e)

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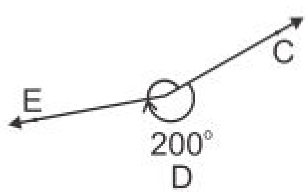
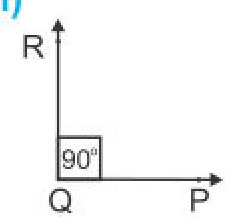


(h)

(i)

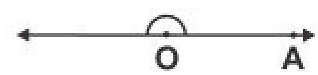


7.1.2 Use protractor to construct a right angle, a straight angle and reflex angles of different measure.

1. To construct a right angle with the help of protractor.

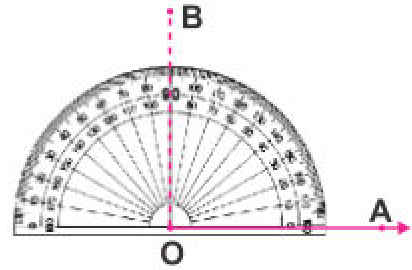
Steps of construction:

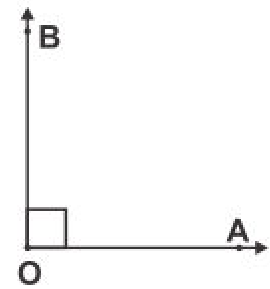
(i) Draw ray $\overrightarrow{OA}$.



(ii) Place the centre point of the protractor at point O along $\overrightarrow{OA}$.

(iii) Read the measurement at the protractor from the side where its zero mark lies on $\overrightarrow{OA}$. Mark the point B at 90°.





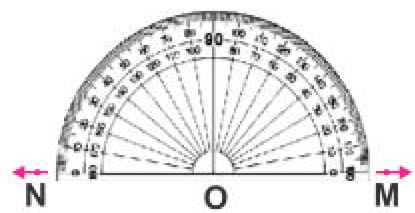
(iv) Draw $\overrightarrow{OB}$ as shown in the picture.
Thus m $\angle$ AOB = 90° and it is a right angle.

2. To construct a straight angle with the help of protractor.

Steps of construction:

(i) Draw a ray $\overrightarrow{OM}$.

(ii) Place the centre point of the protractor on point O along $\overrightarrow{OM}$.



(iii) Read the protractor from the side where its zero mark lies on $\overrightarrow{OM}$. Mark a point N at 180°.

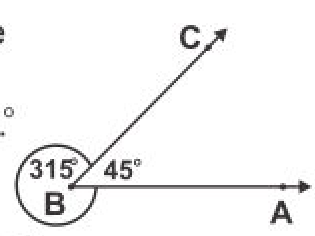
(iv) Now draw $\overrightarrow{ON}$ as shown in the picture.
Thus m $\angle$ MON = 180° which is the required straight angle.

3. Using the protractor to construct reflex angle of different measures.

Steps of construction:

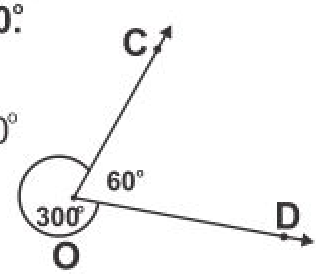
Let us draw and measure a reflex angle of 315° using a protractor.
We know that one complete turn = 360°.

Therefore the measure of the acute angle $\angle ABC = 45^\circ$.
The reflex angle $\angle ABC = 360^\circ - 45^\circ = 315^\circ$.
We first draw and measure the acute $\angle ABC = 45^\circ$.
Then we draw an arc around the outside of



Teacher's Note Teacher should help students to draw different angles step by step.

Example. **Construct a reflex angle of 300°.**



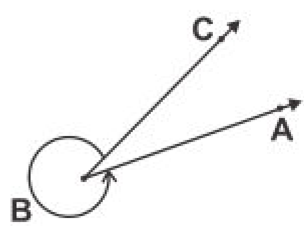
Steps of construction:

We subtract the given measure of 300° from 360° i.e. 360° – 300° = 60°. Now we draw an acute angle of 60° as to get the required reflex angle of 300°. ∠ COD is the required reflex angle.
In this way we can draw a reflex angle of different measure.

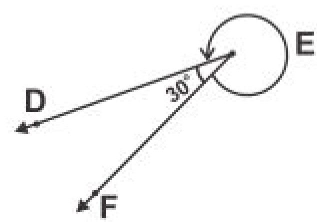
EXERCISE 7.2

1. **Measure the reflex angles by using protractor.**

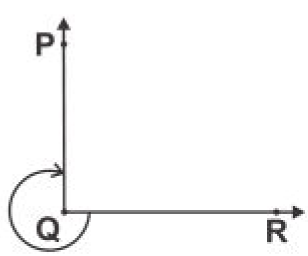
(i)



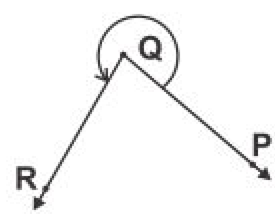
(ii)



(iii)



(iv)



2. **Draw and label the following angles:**

(i) ∠ ABC (reflex angle) (ii) ∠ DEF (straight angle)
(iii) ∠ GHI (reflex angle) (iv) ∠ PQR (right angle)
(v) ∠ STU (straight angle) (vi) ∠ XYZ (reflex angle)

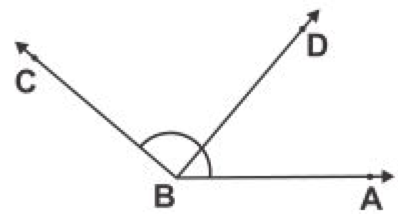
3. **Using protractor, draw the following:**

(i) Reflex ∠ ABC of 310° (ii) Reflex ∠ DEF of 280°
(iii) Reflex ∠ LMN of 340° (iv) Reflex ∠ OPQ of 290°

7.1.3 Describe Adjacent, Complementary and Supplementary angles

We often come across pairs of angles that have some special properties. Some of them are given below:

(i) Adjacent Angles

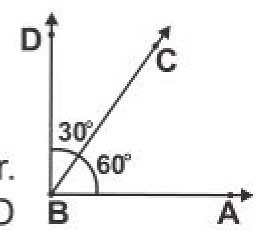


Look at the figure. We have two angles: (i) $\angle ABD$ and (ii) $\angle DBC$. There is a common vertex at point B and a common arm $\overrightarrow{BD}$. The other two arms $\overrightarrow{BA}$ and $\overrightarrow{BC}$ of the angles are on the opposite sides of the common arm $\overrightarrow{BD}$.

Two such angles: $\angle ABD$ and $\angle DBC$ are called adjacent angles.

> Thus two angles are said to be adjacent angles, if they have a common vertex and, a common arm.

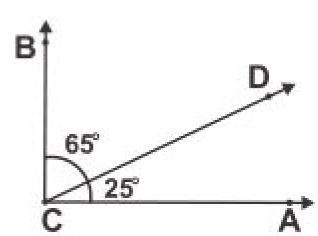
(ii) Complementary Angles



Two angles whose measures have a sum of 90°, are called complementary angles. In figure 60° and 30° are two complimentary angles because $m\angle ABC + m\angle CBD = 60^\circ + 30^\circ = 90^\circ$. Each angle is called a complement of the other. So, $\angle ABC$ is complement of $\angle CBD$ and $\angle CBD$ is complement of $\angle ABC$

Example: Find the complement of 65°.

Solution:

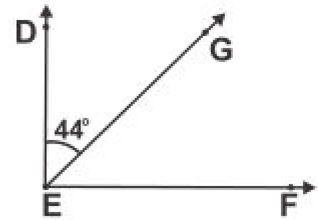


$m\angle BCD$ is 65°, then measure of its complementary angle $\angle ACD$ will be $90^\circ - 65^\circ = 25^\circ$. Because $\angle BCD$ and $\angle ACD$ are two complementary angles.

$m\angle BCD + m\angle ACD = 65^\circ + 25^\circ = 90^\circ$

Teacher's Note Teacher should explain the concept of adjacent, complementary and supplementary angles.

Activity **Find complement of m ∠DEF = 44° in the following figure.**



In figure, ∠DEF and ∠FEG are two complementary angles.

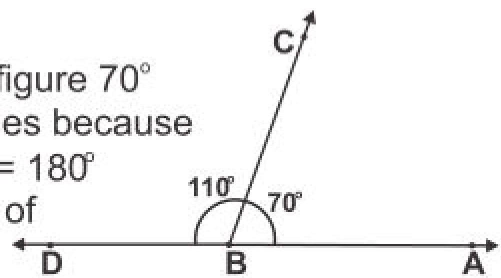
Therefore, m ∠DEF + m ∠FEG = ☐

m ∠DEF = 44°(given)

Hence, m ∠FEG = 90°– 44°= ☐

Thus, the complement of ∠DEF is ☐

(iii) Supplementary Angles



Two angles whose measure have a sum of 180°, are called supplementary angles. In figure 70° and 110° are supplementary angles because m ∠ABC + m ∠CBD = 110°+ 70°= 180° Each angle is called supplement of the other.

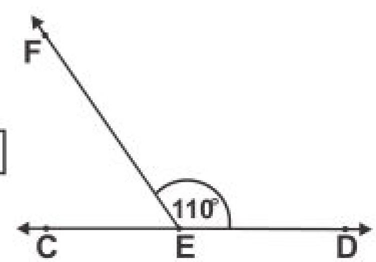
Then ∠ABC is supplement of ∠CBD, or ∠CBD is supplement of ∠ABC.

Example: Find supplement of the given angle of 120°.

Solution:

Supplement of 120°= 180°– 120° = 60°

Activity **Find supplement of the given angle of 110°.**



In figure ∠CEF and ∠DEF are two supplementary angles.

Therefore m ∠CEF + m ∠DEF = ☐

m ∠DEF = 110°(given)

Hence m ∠CEF = 180°– 110°= ☐

Thus the supplement of ∠DEF is ☐

or supplement of 110°is ☐.

EXERCISE 7.3

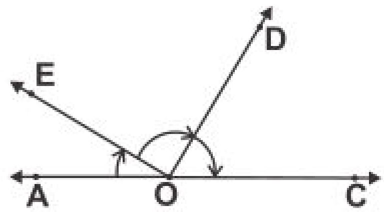
1. Look at the figure and answer the following:

(i) Is $\angle AOE$ adjacent to $\angle DOE$?

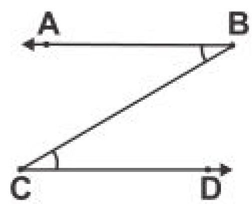
(ii) Is $\angle AOD$ adjacent to $\angle COD$?

(iii) Is $\angle AOE$ adjacent to $\angle AOD$?

(iv) Is $\angle DOE$ adjacent to $\angle EOC$?



2. Is $\angle ABC$ adjacent to $\angle BCD$?
Why or why not?



3. Find the complement of each of the following angles:

(i) 60° (ii) 76° (iii) 45° (iv) 38° (v) 15°

4. Find the supplement of each of the following angles:

(i) 25° (ii) 45° (iii) 70° (iv) 98° (v) 143°

5. Identify the complementary and supplementary pair of angles:

(i) $49^\circ, 41^\circ$ (ii) $154^\circ, 26^\circ$ (iii) $95^\circ, 85^\circ$

(iv) $32^\circ, 58^\circ$ (v) $111^\circ, 69^\circ$ (vi) $14^\circ, 76^\circ$

6. (i) Find the angle which is equal to its complement.
(ii) Find the angle which is equal to its supplement.

7. Can two angles be supplementary, if both of them are:

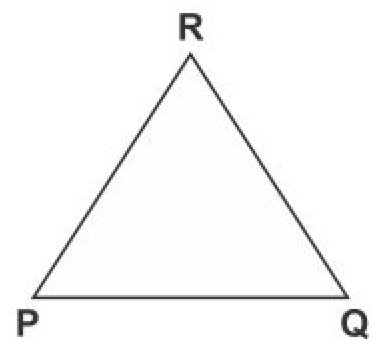
(i) Obtuse (ii) Acute (iii) Right

8. Are complementary angles also adjacent angles? Are supplementary angles also adjacent angles? Explain your answers.

7.2 TRIANGLES

Definition of a triangle:

Triangle is a plane closed figure with three sides. The symbol of triangle is Δ. Given figure is ΔPQR. Points P, Q and R are three vertices. $\overline{PQ}$, $\overline{QR}$ and $\overline{RP}$ are three sides. $\angle PQR$, $\angle QRP$ and $\angle RPQ$ are three angles.



Sum of all three angles of a triangle is equal to 180°.

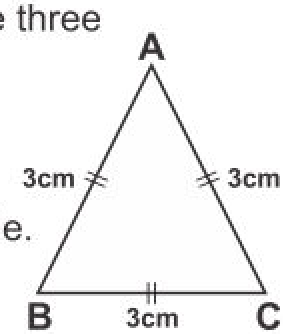
Types of Triangles with respect to their sides

According to the sides of a triangle, there are three types of a triangle.

(i) Equilateral Triangle

It is a triangle having three sides equal in length. ΔABC is an equilateral triangle.

$\therefore$ m $\overline{AB}$ = m$\overline{BC}$ = m $\overline{CA}$

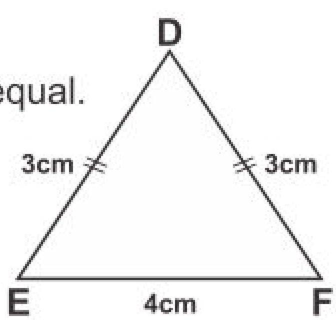


(ii) Isosceles Triangle

It is a triangle having any two sides equal.

ΔDEF is an isosceles triangle.

$\therefore$ m $\overline{DE}$ = m$\overline{DF}$

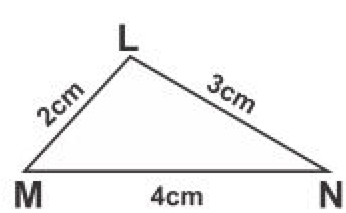


(iii) Scalene Triangle

It is a triangle where all sides are differnts in length.

ΔLMN is scalene triangle.

because m $\overline{LM}$ $\neq$ m $\overline{MN}$ $\neq$ m$\overline{LN}$.



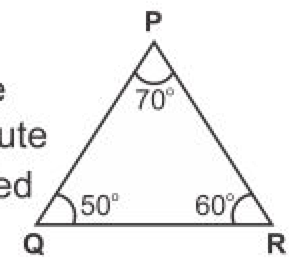
Teacher's Note Teacher should explain the types of triangles with respect to their sides and angles.

Types of Triangles with respect of their angles

According to the angles of a triangle, there are three types of a triangle.

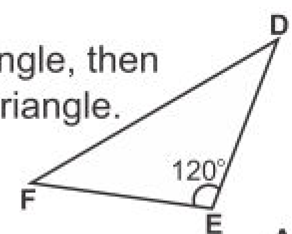
(1) Acute-angled Triangle

If all three angles of a triangle are acute angles, then the triangle is called an acute angled triangle. ΔPQR is an acute angled triangle, in which $\angle P$, $\angle Q$ and $\angle R$ are acute angles.



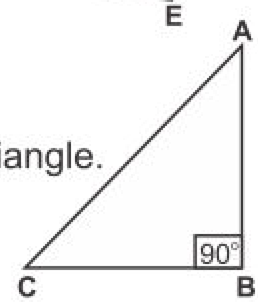
(2) Obtuse-angled Triangle

If one angle of a triangle is an obtuse angle, then the triangle is called an obtuse angled triangle. ΔDEF is an obtuse angled triangle, in which $\angle DEF$ is an obtuse angle.



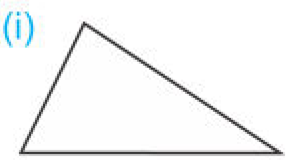
(3) Right-angled Triangle

If one angle of a triangle is a right angle, then the triangle is called a right angled triangle. ΔABC is a right-angled triangle, in which $m\angle ABC = 90^\circ$



EXERCISE 7.4

1. Measure the sides of the following triangles and then write their nameswith respect to their sides.

(i)


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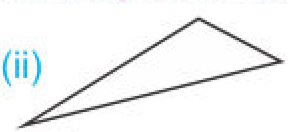

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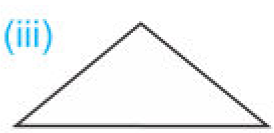

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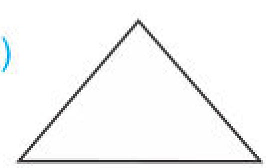

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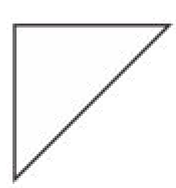

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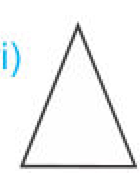
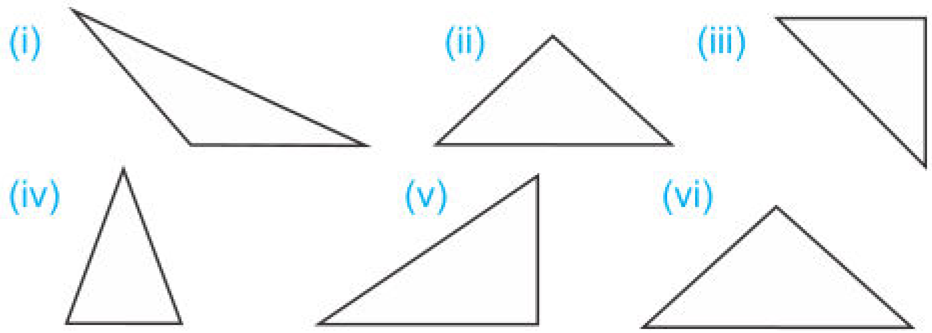


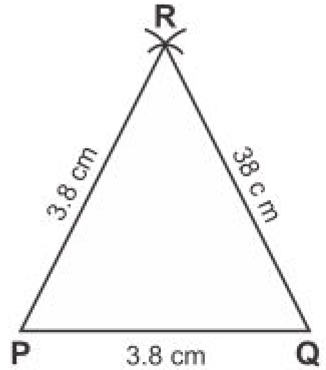
2. Use protractor to measure the angles of the following triangles and write their names with respect to their angles.



7.2.1 Use compasses and straight edge to construct equilateral, isosceles and scalene triangles when three sides are given.

A. Using pair of compasses and ruler to construct an equilateral triangle when the measure of each side is given.

Example 1. Construct a equilateral triangle PQR such that its each side is 3.8 cm.



Given: Each side of triangle is 3.8 cm. Since it is equilateral triangle hence measure of each angles will be 60. °

Steps of construction:

Step 1: Draw a line segment 3.8 cm length with the help of ruler. Name it as $\overline{PQ}$.

Step 2: Take P and Q as centres, open pair of compass to measure 3.8 cm. Draw an arc of radius 3.8 cm from point P end another from point Q. Such as to intersect each other. Mark the point where two arcs meet as R.

Step 3: Draw $\overline{PR}$ and $\overline{QR}$.

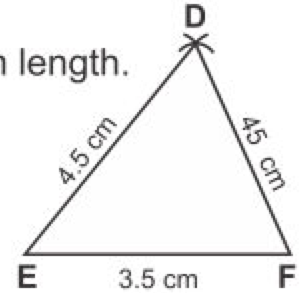
Thus ΔPQR is the required equilateral triangle.

B. Using pair of compasses and ruler to construct an isosceles triangle when measure of sides are given.

Example 2:

Construct a triangle DEF when m $\overline{DE}$ = m $\overline{DF}$ = 4.5 cm and m $\overline{EF}$ = 3.5 cm

Steps of construction:



Step I: Use a ruler and draw $\overline{EF}$ of 3.5 cm length.

Step II: Open the pair of compasses to measure 4.5 cm. Take E and F as centre. Draw two arcs each of 4.5 cm for E and F, such as to intersect other.

Step III: Mark the point where two arcs intersect as D.

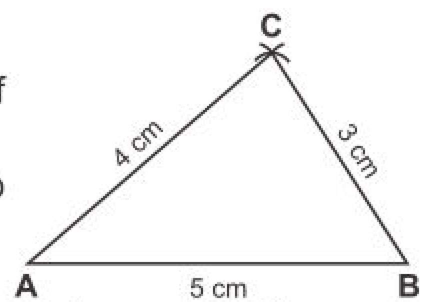
Step IV: Draw $\overline{DE}$ and $\overline{DF}$.

Thus ΔDEF is the required isosceles triangle.

C. To construct a scalene triangle, the measures of all three sides are given. We use the following steps which are given in the example.

Example 3: Construct a triangle ABC when m $\overline{AB}$ = 5 cm, m $\overline{AC}$ = 4 cm and m $\overline{BC}$ = 3 cm

Steps of construction:



Step I: Use a ruler and draw $\overline{AB}$ of 5 cm in length.

Step II: Open pair of compasses to measure 4 cm. Take A as centre and draw an arc.

Step III: Again open pair of compasses to measure 3 cm. Take B as centre and draw another arc to meet previous arc.

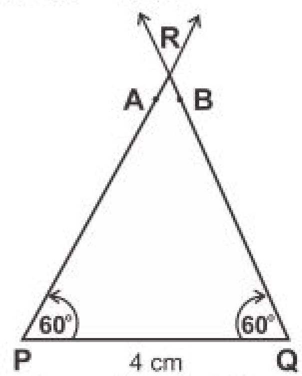
Step IV: Mark the point where two arcs meet; mark as C. Draw $\overline{AC}$ and $\overline{BC}$.

Thus ΔABC is the required scalene triangle.

7.2.2 Use a protractor and ruler to construct equilateral, isosceles and scalene triangles, when two angles and included side are given. Measure the length of the remaining two sides and one angle of the triangle.

We use the following steps given in the examples.

Example 1: Construct an equilateral triangle when
$m\angle PQR = 60^\circ = m\angle QPR$ and $m\overline{PQ} = 4cm$



Steps of construction:

Step I: Draw a line segment PQ measuring 4 cm.

Step II: Place the central point of the protractor at point P. Read from Q and mark point A at 60°.

Step III: Place the central point of the protractor at point Q read from P and mark point B at 60°.

Step IV: From point P and Q segment draw the ray $\overrightarrow{PA}$ and $\overrightarrow{QB}$, so that the rays intersect each other at point R.

Now ΔPQR is the required triangle.

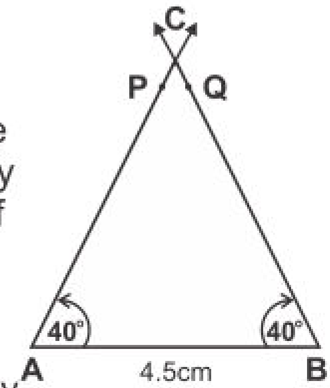
Step V: Measure the sides $\overline{PR}$ and $\overline{QR}$ and $\angle PQR$.

Thus $m\overline{PR} = 4cm$, $m\overline{QR} = 4$ cm and $m\angle PRQ = 60^\circ$
Hence ΔPQR is the required equilateral Δ.

Example 2: Construct an isosceles ΔABC, where $m\overline{AB} = 4.5$ cm,
$m\angle ABC = m\angle CAB = 40^\circ$.

Given: Two angles of equal measured and one side is given.

Steps of construction:



Step I: Draw $\overline{AB}$ of measure 4.5 cm.

Step II: Place the central point of the protractor at point A in such a way that $\overline{AB}$ coincides the base line of the protractor. Read it upto 40° and mark at point P.

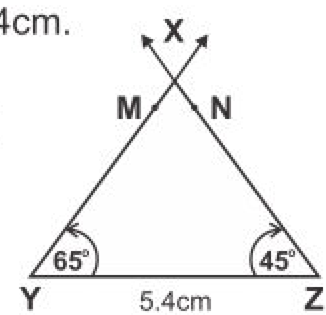
Step III: Place the central point of the protractor at point B in such a way that $\overline{AB}$ coincides the base line of the protractor. Read it upto 40°and mark it point Q.

Step IV: From points A and B, draw $\overrightarrow{AP}$ and $\overrightarrow{BQ}$ to intersect each other at a point.

Step V: Mark the point as C. In this way, we get an isosceles triangle ABC. Now measure the remaining two sides $\overline{AC}$ and $\overline{BC}$ with the help of ruler. Also measure third angle ACB with the help of protractor.

Example 3: Construct a scalene ΔXYZ, when $m\angle XYZ = 65°$
$m\angle XZY = 45°$ and $m\overline{YZ} = 5.4$cm.

Steps of construction:



Step I: With the help of ruler and pencil draw $\overline{YZ}$ of 5.4cm in the length.

Step II: Place protractor at point Y. Read it and mark point upto 65° and mark it as M.

Step III: From point Z, read the protractor up to 45°and mark at a point N.

Step IV: Through points Y and Z, draw $\overrightarrow{YM}$ and $\overrightarrow{ZN}$ to intersect each other at point X.

Step V: Measure $\overline{XY}$, $\overline{XZ}$ and $\angle YXZ$
$m\overline{XY}$ = _____cm, $m\overline{XZ}$ = _____cm and $m\angle YXZ$ = **70°**
Thus ΔXYZ is the required scalene triangle.

7.2.3 Define Hypotenuse of a right angled triangle.

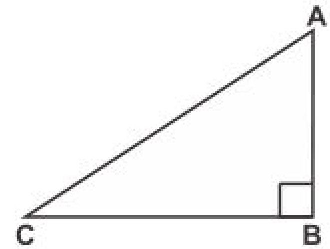
In a right triangle, the side opposite to the right angle is called hypotenuse.

> **Note:** Hypotenuse is the longest side in a right angled triangle.

In figure, ΔABC is a right triangle,

where $\angle B$ is a right angle.

$\overline{AC}$ or $\overline{CA}$ is hypotenuse, as it is opposite to right angle B.



7.2.4 Use protractor and straightedge/ruler to construct a triangle when two angles and included side are given

Example:

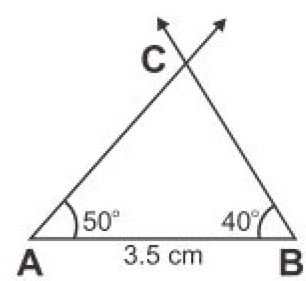
Construct a right angled ΔABC, where $m\angle BAC = 50^\circ$, $m\angle \overline{ABC} = 40^\circ$ and $m\,AB = 3.5$ cm

Steps of construction:

Step I: Draw $\overline{AB}$ of measure 3.5 cm.

Step II: Take point A as centre, draw $\angle BAC$ of measure 50°.

Step III: Take point B as centre, draw $\angle ABC$ of measure 40°.



Step IV: The point where arms of both the angles meet at point C, form right angle.

Thus, ABC is the required right angled triangle.

7.2.5 Use protractor and straightedge/ruler to construct acute angled, obtuse angled and right angled triangles when one angle and adjacent sides are given

Example 1: Construct an acute angled ΔABC, where $m\angle ABC = 75^\circ$, $m\overline{AB} = 4.2$ cm and $m\overline{BC} = 3.8$ cm.

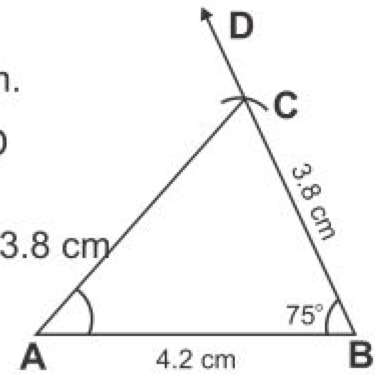
Steps of construction:

Step I: Draw $\overline{AB}$ of measuring 4.2 cm.

Step II: Using protractor make $\angle$ ABD measuring 75°.

Step III: From point B, draw an arc of 3.8 cm to cut the arm $\overrightarrow{BD}$ at point C.

Step IV: Draw $\overline{AC}$. We get the required acute angled ΔABC.

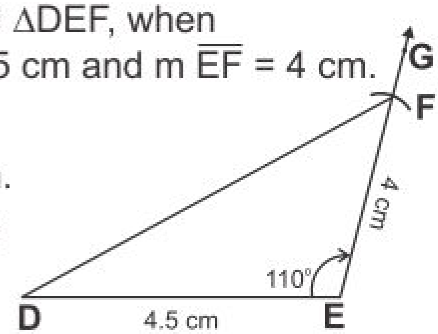


Example 2: Construct an obtuse angled ΔDEF, when m $\angle$ DEF = 110°, m $\overline{DE}$ = 4.5 cm and m $\overline{EF}$ = 4 cm.

Steps of construction:

Step I: Draw $\overline{DE}$ of measuring 4.5 cm.

Step II: Using protractor make $\angle$ DEG measuring 110°.



Step III: Use pair of compass, select E as centre and draw arc of 4 cm radius to cut the arc $\overrightarrow{EG}$ at point F. Use scale and draw $\overline{DF}$, which is the third side of the ΔDEF.

Step IV: Thus, we get the required obtuse angled ΔDEF.

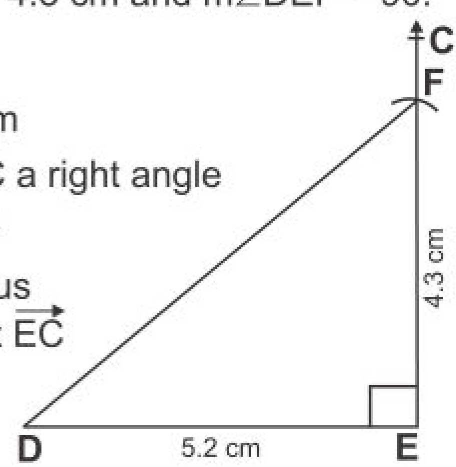
Example 3: Construct a right angled triangle with sides m$\overline{DE}$ = 5.2 cm, m$\overline{EF}$ = 4.3 cm and m$\angle$DEF = 90°.

Steps of construction:

Step I: Draw $\overline{DE}$ measuring 5.2cm

Step II: From point E, draw $\angle$DEC a right angle with the help of protractor.

Step III: With E as centre and radius 4.3 cm, draw an arc to cut $\overrightarrow{EC}$ at point F, the third vertex of ΔDEF.



Step IV: Draw $\overline{DF}$ by joining the points D and F.

Thus ΔDEF is the required right angled triangle.

EXERCISE 7.5

1. Using a ruler and a pair of compasses, construct the following equilateral triangles.

(i) ΔABC where $m\overline{AB} = m\overline{BC} = m\overline{CA} = 4$ cm.

(ii) ΔDEF where $m\overline{DE} = m\overline{EF} = m\overline{DF} = 3.5$ cm.

(iii) ΔPQR where $m\overline{PQ} = m\overline{QR} = m\overline{PR} = 5.2$ cm.

2. Using a ruler and a pair of compasses, construct the following isosceles triangles.

(i) ΔABC where $m\overline{AB} = m\overline{BC} = 6$ cm; $m\overline{AC} = 4$ cm.

(ii) ΔDEF where $m\overline{DE} = 2.5$ cm, $m\overline{EF} = m\overline{DF} = 4$ cm.

(iii) ΔPQR where $m\overline{PQ} = 4$ cm, $m\overline{QR} = m\overline{PR} = 3.5$ cm.

3. Using a ruler and a pair of compasses, construct the following scalene triangles.

(i) ΔABC where $m\overline{AB} = 4.8$ cm, $m\overline{BC} = 3$ cm and $m\overline{AC} = 5$ cm.

(ii) ΔPQR where $m\overline{PQ} = 4.5$ cm, $m\overline{QR} = 5$ cm and $m\overline{PR} = 3.5$ cm.

(iii) ΔEFG where $m\overline{EF} = 5.2$ cm, $m\overline{FG} = 4.4$ cm and $m\overline{GE} = 3$ cm.

4. Using ruler, protractor and a pencil to construct the following equilateral, isosceles and scalene triangles. Also measure the remaining two sides and one angle in each triangle.

(i) An equilateral ΔABC where $m\overline{AB} = 5.7$ cm,
$m\angle ABC = m\angle BAC = 60^\circ$.

(ii) An isosceles ΔLMN where $m\overline{LM} = 5$ cm, $m\angle LMN = m\angle MLN = 70°$.

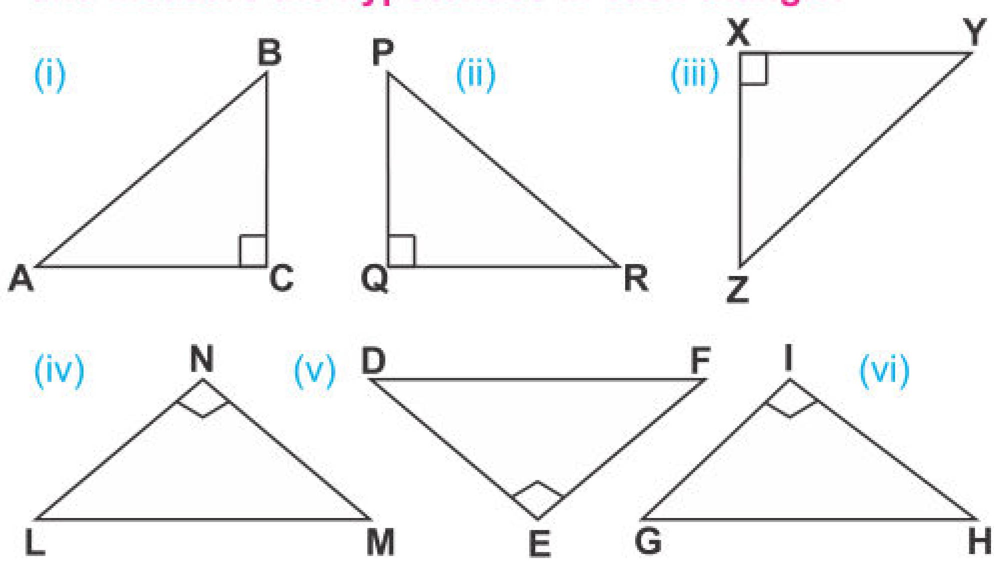
(iii) A scalene ΔXYZ where $m\overline{XY} = 6$ cm, $m\angle XYZ = 60°$, $m\angle YXZ = 50°$.

(iv) An equilateral ΔRST where $m\overline{RS} = 4.8$ cm and $m\angle RST = m\angle TRS = 60°$.

(v) A scalene ΔEFG where $m\overline{EF} = 5.5$ cm, $m\angle EFG = 75°$ and $m\angle GEF = 65°$.

(vi) An isosceles ΔJKL where $m\angle JLK = 45°$, $m\overline{JL} = 4.6$ cm and $m\angle KJL = 45°$.

5. Look at the following right angled triangles. Name and measure the hypotenuse in each triangle.



6. Using a ruler and protractor to construct the following triangles. Measure the length of remaining two sides and one angle of the triangle.

(i) ΔABC where
$m\overline{AB} = 5$ cm,
$m\angle BAC = 55°$,
$m\angle ABC = 35°$

(ii) ΔJKL where
$m\angle KJL = 65°$,
$m\angle JKL = 25°$,
$m\overline{JK} = 4.8$ cm

(iii) ΔPQR where
$m\angle QPR = 30^\circ$
$m\angle PQR = 60^\circ$,
$m\overline{QP} = 4$ cm

(iv) ΔSTU where
$m\overline{ST} = 5.3$ cm,
$m\angle STU = 75^\circ$
$m\angle TSU = 15^\circ$

7. Use protractor, a pair of compasses and ruler to construct the following triangles:

(i) Acute angled ΔABC, when $m\angle BAC = 65^\circ$, $m\overline{AB} = 3.6$ cm and $m\overline{AC} = 4.4$ cm.

(ii) Right angled ΔDEF, when $m\angle DEF = 90^\circ$, $m\overline{DF} = 3$ cm and $m\overline{EF} = 4$ cm.

(iii) Obtuse angled ΔLMN, $m\angle NML = 110^\circ$, $m\overline{MN} = m\overline{ML} = 5$ cm.

(iv) Acute angled ΔPQR, $m\angle PQR = 60^\circ$, $m\overline{PQ} = m\overline{QR} = 4$ cm.

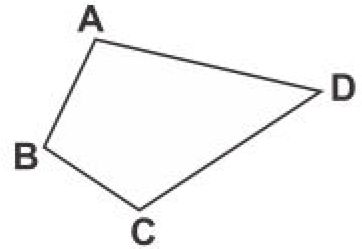
(v) Right angled ΔXYZ, when $m\angle XYZ = 90^\circ$, $m\overline{XY} = 4.2$ cm $= m\overline{YZ}$

(vi) Obtuse angled ΔSTU, when $m\angle STU = 120^\circ$, $m\overline{ST} = 3.6$ cm and $m\overline{TU} = 4.2$ cm.

7.3 QUADRILATERALS

A closed figure with four sides is called a quadrilateral

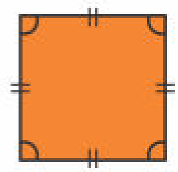
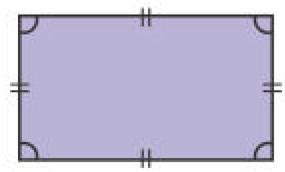
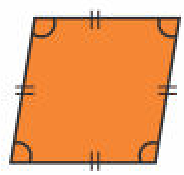
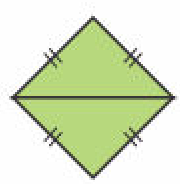
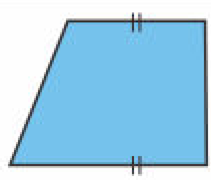
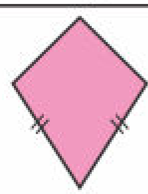
In the figure, quadrilateral ABCD has four sides $\overline{AB}$, $\overline{BC}$, $\overline{CD}$ and $\overline{AD}$. It has four angles $\angle A$, $\angle B$, $\angle C$ and $\angle D$.



The sum of measure all four angles of quadrilateral is equal to 360°.

7.3.1 Recognize the kinds of quadrilateral (square, rectangle, parallelogram, rhombus, trapezium and kite).

The kinds of a quadrilateral are:

Quadrilateral	Figure	Properties
Square		▪ All four sides are equal ▪ Opposite sides parallel ▪ Each angle is of 90°
Rectangle		▪ Opposite sides equal ▪ Opposite sides are parallel ▪ Each angle is of 90°
Parallelogram		▪ Opposite sides equal ▪ Opposite sides are parallel ▪ Opposite angles are equal. ▪ None of angle measure 90°
Rhombus		▪ Four equal sides ▪ Opposite sides are parallel ▪ Opposite angles are equal. ▪ None of angle measure 90°
Trapezium		▪ Only one pair of opposite parallel sides
Kite		▪ Two pairs of adjacent equal sides ▪ Here one pair of equal angles

Teacher's Note Teacher should help students to recognize the kinds of quadrilateral.